

Dhivyadharshini Karunakaran

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PROFESSIONAL SUMMARY

Data Analyst with an MSc in Computer Science (Data Analytics) and IEEE-published research, experienced in transforming complex data into actionable business insights through SQL, Python, Power BI, ETL, and data visualization. Skilled in data cleaning, feature engineering, reporting, dashboard development, and building data-driven solutions to support business decision-making and process improvement. A proactive problem solver with strong analytical and communication skills, committed to delivering accurate, scalable, and impactful data solutions. Stamp 1G visa holder, fully eligible to work full-time in Ireland and available for immediate start.

WORK EXPERIENCE

Graduate Teaching Assistant (CT2102)

Oct '24 — May '25

University of Galway

Galway, Ireland

- **Role:** Delivered practical Java and Web Application Development labs, supporting students with object-oriented programming, application logic, front-end development, debugging, code review, and troubleshooting technical issues during development.
- **Tools & Outcome:** Applied Java, HTML, CSS, JavaScript, Git, and Moodle to diagnose coding errors, test and debug applications, review implementations, manage coursework workflows, and provide structured technical guidance while maintaining consistent assessment standards.

Freelance Web Developer

Nov '23 — May '25

Freelance

Galway, Ireland

- **Role:** Engineered responsive web applications from requirements through deployment, translating client requirements into application functionality, developing front-end components, integrating database-driven features, and managing testing, debugging, deployment, and maintenance.
- **Tools & Outcome:** Applied HTML, CSS, JavaScript, Bootstrap, PHP, MySQL, Git, and GitHub to build responsive interfaces, implement application logic, execute SQL queries, manage relational data, troubleshoot defects, optimize application performance, and maintain version-controlled codebases.

Data Intern

Jan '23 — Feb '24

8Queens Software Technologies Pvt. Ltd.

Tamil Nadu, India

- **Role:** Executed end-to-end data preparation and analysis workflows, including data collection, cleaning, transformation, validation, exploratory analysis, and preparation of structured datasets for reporting and business analysis.
- Applied Python and SQL for data manipulation, querying, transformation, and validation; used Excel and Google Sheets for reconciliation, quality checks, analysis, and reporting, while maintaining technical documentation and version-controlled analytical work through GitHub.

Java Intern

Feb '22 — Dec '22

Paxel Techno Labs

India

- **Role:** Developed and maintained Java application components, implementing application functionality, integrating database operations, debugging defects, testing software behaviour, and collaborating on application enhancements and technical issue resolution.
- **Tools & Outcome:** Applied Java, MySQL, Git, and Eclipse to implement application modules, execute database operations, perform debugging and functional testing, manage source-code changes through version control, and document implementations to improve application reliability and maintainability.

EDUCATION

MSc in Computer Science (Data Analytics), University of Galway

'24 — '25

Galway

B.Tech in Computer Science and Business Systems, Sri Sairam Engineering College

'20 — '24

Tamil Nadu, India

TECHNICAL SKILLS

Skills SQL, Python (Pandas, NumPy), Microsoft Power BI (DAX, Power Query), Microsoft Excel, Data Visualization, Exploratory Data Analysis (EDA), KPI Reporting, ETL & Data Pipelines, Data Warehousing, Dimensional Data Modelling, Data Cleaning & Transformation, Data Validation & Data Quality, Machine Learning, Scikit-learn, Natural Language Processing (NLP), Feature Engineering, Model Evaluation, SQL Server, MySQL, Flask, Git, GitHub, Java, HTML, CSS, JavaScript, Microsoft Azure, Software Testing, Computer Networks & Protocols, Cybersecurity Fundamentals

RESEARCH PUBLICATIONS

Machine Learning-Based Tamil Sign Language Recognition Tool for the Deaf and Hard-of-Hearing Community

IEEE, Scopus Indexed

Assessment of Student Attentiveness in Classroom Environment Using Deep Learning

IEEE

Roll Call Tracking System for Education Sector Using Face Recognition Systems

IEEE

PROJECTS

Investment Portfolio Analytics & Reporting Platform

- Role: Designed and developed an end-to-end investment analytics platform to manage portfolio and fund performance data. Responsible for SQL database design, data modelling, ETL processes, reporting automation, and business intelligence development to support investment performance analysis.
- **Tools & Outcome:** Used SQL for relational data modelling and analytical queries, Python (Pandas) for ETL and data validation, and Power BI (Power Query, DAX) to develop dashboards tracking ROI, asset allocation, investment trends, and fund performance.

Financial Data Warehouse & Automated ETL Pipeline

- Role: Developed a centralized financial data warehouse by integrating data from multiple business sources. Led data extraction, transformation, loading (ETL), and dimensional modelling to improve reporting efficiency and support business decision-making.
- **Tools & Outcome:** Used SQL Server for dimensional modelling and reporting queries, Python (Pandas) for automated ETL and data cleansing, and Power BI (Power Query, DAX) to visualize executive KPIs, financial metrics, and operational performance.

Fund Accounting Reconciliation & Data Quality System

- Role: Built a reconciliation and data quality solution to validate fund accounting transactions and financial records across multiple datasets. Responsible for developing reconciliation logic, exception reporting, and data validation processes to improve reporting accuracy.
- **Tools & Outcome:** Used SQL for transaction matching and reconciliation queries, Python (Pandas) for validation and exception detection, and Power BI to monitor NAV, cash movements, duplicate/missing records, reconciliation status, and data-quality KPIs.

Aer Lingus Flight Network Analysis

- Role: Conducted network analysis on Aer Lingus flight route data to identify connectivity patterns, optimize route insights, and support data-driven decision-making. Responsible for data cleaning, SQL-based data preparation, feature engineering, and exploratory analysis.
- **Tools & Outcome:** Used SQL for route-data preparation, Python (Pandas, NetworkX) for feature engineering and graph analysis, applying degree and betweenness centrality to identify key routes and airports, with Power BI dashboards visualizing network connectivity and performance.

Smart Healthcare Assistant (NLP Chatbot)

- Role: Developed an AI-powered healthcare assistant to provide symptom assessment, health guidance, and appointment recommendations. Responsible for NLP pipeline development, data preprocessing, intent recognition, conversational workflow design, and business requirement analysis.
- **Tools & Outcome:** Used Python and NLP for text preprocessing, tokenization, lemmatization and intent classification, machine learning for response prediction, Flask for application integration, and SQL for structured data storage and retrieval.

CERTIFICATIONS

Mayerfeld Practicum Program – Data Analyst | Google Data Analytics Professional Certificate | Microsoft Azure Data Fundamentals (DP-900) | Oracle Java Fundamentals | Machine Learning Model Development Using Flask | Python for Data Science | SQL for Data Analysis | Cyber Security | Software Testing | Computer Networks & Protocols | Data Warehousing Fundamentals | ETL & Data Pipelines